

ARTIN (ALIREZA) KAVOOSI M.Sc.

EDUCATION	University of Tehran Tehran, Iran <i>M.Sc. in System Modeling and Data Analytics</i> Sep 2023 - Present - GPA: 18.62/20, (4/4) - Supervisor: Prof. Tavakkoli-Moghaddam - Thesis: “Towards Autonomous Supply Chains for Perishable Products: Learning-Based Decision-Making for Coordination, Competition, and Resilience” (expected completion: Sep 2026)
	Kharazmi University Tehran, Iran <i>B.S. in Industrial Engineering</i> Sep 2019 - Jun 2023 GPA: 16.44/20
RESEARCH INTERESTS	Methods: Markov Decision Processes, Linear & Integer Programming, Robust Optimization, Bandit Algorithms. Domain: Healthcare Operations, Inventory Control, Supply Chain Management
PUBLICATIONS	Published Papers - Kavoosi, A., Tavakkoli-Moghaddam, R., Sajedi, H., Tajik, N., & Tafakkori, K. (2025) Dynamic Pricing and Inventory Control of Perishable Products by a Deep Reinforcement Learning Algorithm. <i>Expert Systems with Applications</i>, 291, 128570 [DOI]. Preprint Papers - Kavoosi, A. (2026) Stochastic Queens Elimination. Available online (SSRN). [Project page and code] Working and Under Review Papers - Kavoosi, A., Tavakkoli-Moghaddam, R., Ziari, M., & Azaron, A. Coordinating Pricing and Replenishment in a Blood Supply Chain with Deep Reinforcement Learning. Under review. - Kavoosi, A., Tavakkoli-Moghaddam, R. Disruption-Resilient Blood Supply Chain by a Structural Causal Graph Dynamics and Decentralized Reinforcement Learning Approach. Under review. - Kavoosi, A., Tavakkoli-Moghaddam, R., Dolgui, A. Cross-Entropy Neural Networks for Mixed Strategies in a Finite-Horizon Supplier–Retailer Inventory Duopoly with Lead Times and Rationing. Submitted. - Kavoosi, A., Tavakkoli-Moghaddam, R. Online Learning of Neural Ordinary Differential Equations for Dynamic Inventory Control under Non-Stationary Demand. Submitted.
AWARDS AND HONORS	Top 0.4% among +10,000, University Master’s Degree Entrance Exam (Konkour) in Iran, Industrial Engineering 2023 Top 1% among +150,000, Nationwide University Entrance Exam (Konkour) in Iran, Mathematics and Physics 2019

ACADEMIC EXPERIENCE	Research Assistant University of Tehran Sep 2024 - Present Studying and researching on inventory control and supply chain problems, supervised by Prof. Tavakkoli-Moghaddam.	
	Teaching Assistant University of Tehran Mar 2024 - Sep 2025 - Machine Learning (Spring 2025, Instructor: Prof. Jamili) - Macroeconomics (Spring 2025, Instructor: Prof. Ghaderi) - Soft Computing (Fall 2024, Instructor: Prof. Tavakkoli-Moghaddam) - Linear Algebra (Spring 2024, Instructor: Prof. Rahbari)	
	Teaching Assistant Kharazmi University Mar 2023 - Aug 2023 Engineering Mathematics II (Spring 2023, Instructor: Prof. Khamse)	
WORK EXPERIENCE	Business Partner Family Fashion Boutiques (Privately Owned) Jun 2020 - Present - Co-manage three family-owned fashion boutiques (Aylar & Tirajeh) serving customers across Tehran Province, Iran. - Oversee daily store operations, including inventory management, merchandising, and sales activities. - Deliver high-quality customer service while supporting business growth and maintaining long-term customer relationships.	
	Research Intern Sapco Company Apr 2022 - Oct 2022 - On data-driven transportation cost estimation, application of SAP tools and software to optimize inventory operations, and analyzing workflows using BPMN. - SAPCO is an Iranian automotive parts supply and procurement company that manages the sourcing and quality of components for Iran Khodro, one of Iran's largest automotive manufacturers.	
SERVICE	Reviewer for <i>Transportation Research Part E: Logistics and Transportation Review</i>	
SKILLS	Programming Languages:: Python, Julia, GAMS, IBM SPSS, LaTeX. Research Software Portfolio: Develop open-source implementations of optimization algorithms, reinforcement learning methods, scientific visualization, and computational experiments. github.com/akavosi	
LANGUAGES	Persian (Native), Turkish (Native), English (Professional).	
REFERENCES	Prof. Reza Tavakkoli-Moghaddam School of Industrial Engineering University of Tehran tavakoli@ut.ac.ir	Prof. Seyed Farid Ghaderi School of Industrial Engineering University of Tehran ghaderi@ut.ac.ir
	Prof. Nazanin Tajik Department of Industrial and Systems Engineering Mississippi State University tajik@ise.msstate.edu	Prof. Matineh Ziari School of Industrial Engineering University of Tehran m.ziari@ut.ac.ir